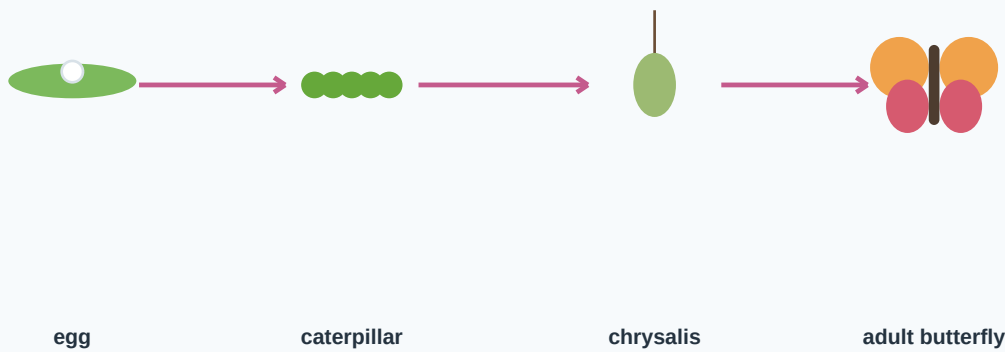


Butterfly Metamorphosis: Four Amazing Stages

See how a tiny egg becomes a winged adult



Metamorphosis changes the animal's body at each stage

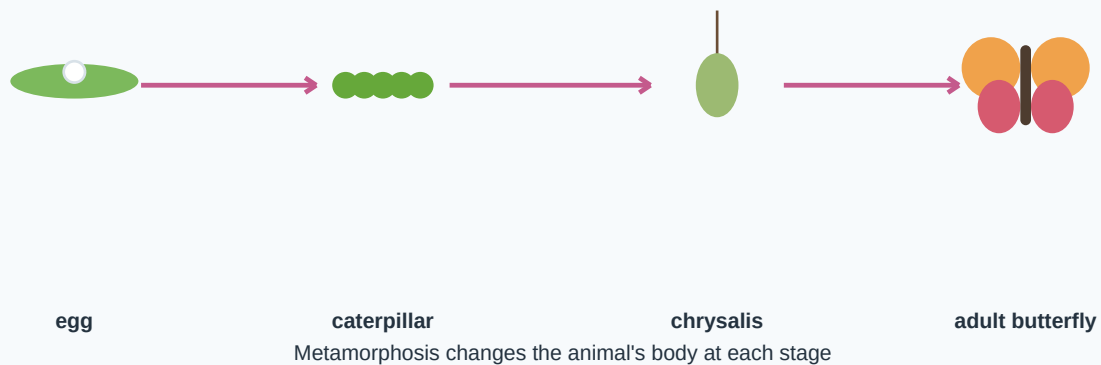
Recommended reading level: Ages 7-11

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What you will learn

- | | |
|---|--|
| 1 | Name the four stages of a butterfly's life cycle. |
| 2 | Explain what happens inside a chrysalis. |
| 3 | Describe how caterpillars and adult butterflies have different jobs. |
| 4 | Understand why suitable habitats are needed at every stage. |

Key words: life cycle, egg, larva, caterpillar, chrysalis, pupa, metamorphosis, adult



1. A life of complete change

A butterfly does not begin life with wings. It passes through four very different stages: egg, larva, pupa and adult. This complete change in body form is called **metamorphosis**.

Each stage has a special purpose. The young caterpillar mainly eats and grows. The adult butterfly mainly travels, finds a mate and lays eggs for the next generation.

The four stages form a cycle because adult butterflies lay the eggs that begin the process again.

2. Stage one: the egg

A female butterfly usually lays eggs on a plant that the young caterpillar can eat. Different butterfly species need different host plants. An egg may be round, oval or ribbed and is often smaller than a grain of rice.

Inside the egg, an embryo develops. When it is ready, a tiny caterpillar cuts or chews its way out.

Visual moment to look for: a close-up of butterfly eggs attached to the underside of a leaf.

3. Stage two: the caterpillar

The caterpillar is the larval stage. Its main task is to eat. As it grows, its outer skin becomes too tight. It sheds this skin several times in a process called moulting.

Caterpillars store energy and nutrients for the next stage. Some have colours or markings that help them hide, while others warn predators that they taste unpleasant.

A caterpillar may become many times larger between hatching and forming a chrysalis.

4. Stage three: the chrysalis

When fully grown, the caterpillar finds a safe place and forms a protective case called a **chrysalis**. The animal inside is called a pupa.

Inside, major changes take place. Many larval tissues are reorganised, and adult structures such as wings, long legs and compound eyes develop. The pupa may look still, but its body is extremely active.

Visual moment to look for: time-lapse footage of a caterpillar becoming a chrysalis and an adult emerging.

5. Stage four: the adult butterfly

The adult slowly pushes out of the chrysalis. At first, its wings are soft and folded. It pumps fluid into them and waits while they expand and harden.

Adult butterflies drink nectar using a long tube called a proboscis. As they visit flowers, some carry pollen between plants. After mating, females lay eggs and the life cycle begins again.

Butterflies need host plants for caterpillars and nectar plants for adults. A healthy habitat provides both.

Chapter summary

- Butterflies pass through egg, larva, pupa and adult stages.
- The caterpillar eats and stores energy for transformation.
- Inside the chrysalis, the body is reorganised into the adult form.
- Adult butterflies feed, reproduce and begin the next generation.

Glossary

Word	Meaning
metamorphosis	A major change in body form during an animal's life cycle.
larva	The young feeding stage of an animal such as a butterfly.
caterpillar	The larva of a butterfly or moth.
pupa	The transformation stage between larva and adult.
chrysalis	The protective pupa covering of a butterfly.
proboscis	A long tube used by a butterfly to drink liquids.

Quick check

1. What are the four stages of a butterfly's life cycle?
2. Why are eggs often laid on a particular host plant?
3. What is the main job of a caterpillar?
4. What changes happen inside a chrysalis?
5. What must an adult butterfly do before it can fly?

Testing note: This chapter is intentionally written with clear concepts, visual events and child-friendly language. It is suitable for testing chapter extraction, concept detection, media search, narration, quiz generation and revision-video workflows.